2026 Net Zero Tech International Contest

Pump-Free Single-Phase Immersion Cooling Combined with Heat Pipes

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1. Abstract

This proposal aims to deal with the pain points of traditional single-phase immersion cooling, which relies on water pumps and causes localized heat accumulation at low flow rates. To improve this, we propose a low-energy, pump-free cooling design combining finned heat pipes with an internal circulation fan.

For the practical testing design, we will use a 100W heating plate to simulate the heat generation of high-power electronic components, using EC-110 cooling fluid. We decided to simulate first to estimate the design feasibility. By using COMSOL Multiphysics, we will build models to predict the fluid and temperature distribution. Subsequently, we will move to the practical parts by building the acrylic cooling tank to run controlled experiments. We will compare setups with and without finned heat pipes, as well as varying internal circulation fan powers work in 50% and 100%.

Through this cross-verifying process, we expect to prove that the internal circulation fan can effectively break the thermal boundary layer. Our main goal is to bring away the 100W heat load and allow the system to reach thermal equilibrium through the phase-change heat transfer of the heat pipes, completely without external water pumps. Finally, we will cross-verify the measured steady-state temperature data against the previous COMSOL results to evaluate the accuracy of our physical model. This design will prevent temperature surging, minimize non-computing power consumption to keep the PUE low, and allow us to test different situations or improve the system without much cost.

2. Motivation and Purpose

With the boosting of AI, the pursuit of computing power is continuously surging, bringing massive energy consumption. Since electrical power cannot be fully used for computing, the remaining energy dissipates in thermal form. Currently, standard servers use fans to generate external forced convection to bring away the accumulated heat. However, traditional air cooling can no longer deal with the surging energy density. It is estimated that by 2030, global data center electricity consumption will exceed 1,000 TWh, with cooling systems accounting for approximately 40%, resulting in a Power Usage Effectiveness (PUE) of around 1.6 [1]. The PUE formula is as follows:

$$PUE = \frac{Total\ Facility\ Energy}{IT\ Epuipment\ Energy} \quad \text{Eq.(2.1)}$$

Therefore, we will design a low-energy single-phase immersion cooling (SPIC) system. Based on traditional single-phase immersion cooling, we combine heat pipes and use

COMSOL Multiphysics as the simulation platform. Under the boundary condition of external forced convection, we simulate the heat accumulation of the CPU operating in the dielectric fluid. Ultimately, we aim to keep the PUE below the 1.3 standard set by the Ministry of Economic Affairs to ensure optimal energy efficiency.

3. Literature Review

3.1 COMSOL CFD Simulation:

3.1.1 Cooling tank

To reduce the PUE, Huang, Y. P. et al. [3] in 2023 utilized the ANSYS 2021 ICEM CFD software package to investigate the convection conditions in the cooling tank using three different convection models. These models are: (a) Mixed convection operating mode (MCOM), which simultaneously considers convection caused by the pump and gravity; (b) Virtual convection operating mode (VCOM), which only considers the pump-driven convection while ignoring gravity; and (c) Natural convection operating mode (NCOM), which only considers natural convection driven by temperature and density differences.

The temperature distributions under these different conditions are shown in **Fig. 1**. The maximum CPU surface temperature in the NCOM system was 66.6% and 52.8% higher than those in the MCOM and VCOM systems, respectively. In the subsequent evaluation of energy efficiency levels, the study compared the calculated PUE of the systems when the maximum CPU surface temperature reached 90°C. The results demonstrated that the PUE of the pump-driven SPIC was 1.07, whereas the PUE of the natural convection-driven SPIC was 1.21. Therefore, this study will adopt a forced convection system as the simulation condition for the

heat exchange between the CPU and the coolant.

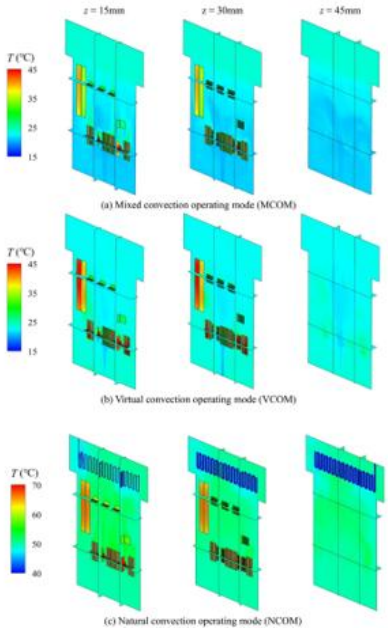


Fig.1 Temperature Distributions Modes of the Three Convection



Fig.2 Axial Fan Installation Schematic

3.1.2 Internal Flow Disturbance and Thermal Non-uniformity Issues

Compared to traditional SPIC systems, this study reduces the cooling system's energy consumption by eliminating the external water pump and modifying the system into an all-in-one integrated design. However, due to the lack of external forced convection typically generated by an external pump drawing fluid in and out of the tank, heat accumulation will occur on the surface of the working components if internal flow disruption is not implemented.

In 2024, Bo Ding et al. [4] proposed utilizing an axial fan in the middle of the cooling tank to draw lower-temperature coolant from both sides of the Server Motherboard (SMB) toward the center, thereby enhancing flow disruption and breaking the thermal boundary layer. The installation position is shown in Fig. 4. The results indicated that the surface temperature of the upstream CPU decreased by 4.21% to 4.96%, whereas that of the downstream CPU only decreased by 1.11% to 1.27%. The local Nusselt number of the downstream CPU increased by 4.04% to 5.80%.

$$Nu = \frac{hl}{k} \qquad Eq.(3.1)$$

In Eq.(3.1), we can see the relation between the Nusselt number(Nu) and the heat convection coefficient(h). It is proven that the fan effectively enhances the local convective heat transfer coefficient. Furthermore, the Delta-AFB0912VH fan used in the literature has a

very low power consumption, and its disruption range is confined to the installation area. This allows for targeted flow agitation specifically in the heat exchange region with minimal energy consumption, aligning with this study's objective of lowering the PUE.

3.1.3 Cooling Tank Boundary Conditions

The cooling tank designed in this study will reference the fluid field assumptions from the 2026 study by Zhou, Y. et al. [2], which use ANSYS Fluent 2020 R2 to simulate SPIC systems. We will use these assumptions to calculate the fluid field after agitation by the internal circulation fan. The assumptions are as follows: (1) incompressible fluid; (2) constant physical properties; (3) three-dimensional flow; and (4) steady-state flow. For the coolant, we will use EC-110 developed by Engineered Fluids, which has a Global Warming Potential (GWP) of 0, aligning with our goal to mitigate global warming.

The turbulence model will employ the computationally efficient $RNG\ k-\varepsilon$model. To compensate for its potential errors and obtain a more accurate steady-state flow distribution, the residual convergence criterion will be set to 10^{-4}.

3.1.4 Heat Pipes

Since the working principles of heat pipes involve phase change and complex thermodynamic models, it is challenging to achieve high simulation accuracy while maintaining reasonable computational costs. According to Joshua E. Hansel et al. [6] in 2025, heat pipe simulation methods can be categorized into the following three approaches: (1) using simple and highly efficient algorithms; however, due to insufficient spatial resolution, this model is unsuitable for coupled reactor analysis. (2) Employing a thermal conduction approximation to simulate heat pipes by setting the effective thermal conductivity to an arbitrarily large value. Compared to models involving phase change, this significantly reduces computational costs and yields a more robust solver. (3) Utilizing high-fidelity heat pipe models to simulate the internal working fluid flow and the heat conduction of the pipe wall. The advantage of this approach lies in its ability to capture the limiting behaviors of heat pipes under real-world conditions.

In their research, Joshua E. Hansel et al. referenced the Liquid-Conduction, Vapor-Flow (LCVF) model proposed by Hansel et al. [7] in 2024. This model applies two-dimensional heat conduction to the heat pipe envelope, liquid layer, and porous media region, while treating the vapor phase as a one-dimensional compressible fluid. By solely considering the flow dynamics of the vapor phase, it resolves the severe convergence issues typically encountered by two-phase flow models during startup conditions, and it utilizes a first-order Lagrangian finite element analysis method. This model has been verified through steady-state and limiting-case analytical solutions, demonstrating its ability to accurately predict heat pipe

temperatures under both steady-state and transient operations. The vapor flow equations used in this study are as follows:

$$\frac{\partial \rho_v A_v}{\partial t} + \frac{\partial \rho_v A_v u_v}{\partial x} = \Gamma_{int} A'_{int} \quad \text{Eq. (1.8)}$$

$$\frac{\partial \rho_v A_v u_v}{\partial t} + \frac{\partial (\rho_v u_v^2 + p_v) A_v}{\partial x} = -F_v A_v + \rho_v g_v A_v \quad \text{Eq. (1.9)}$$

$$\frac{\partial \rho_v E_v A_v}{\partial t} + \frac{\partial u_v (\rho_v E_v + p_v) A_v}{\partial x} = \rho_v u_v g_x A_v + (q_v^{int} + \Gamma_{int} E_v^{int}) A'_{int} \quad \text{Eq. (1.10)}$$

Equations (1.8), (1.9), and (1.10) describe the conservation of mass, momentum, and energy, respectively. In Eq. (1.8), ρ_v denotes the vapor density, A_v represents the cross-sectional area of the vapor channel, u_v is the vapor flow velocity, and Γ_{int} is the phase-change mass flux at the liquid-vapor interface, which describes the evaporation and condensation processes. Additionally, A'_{int} indicates the interfacial area per unit length, representing the contact surface area between the liquid and vapor phases.

Equation (1.9) inherits the definitions from Eq. (1.8), where F_v is the momentum source term accounting for wall friction and porous media resistance, and $g_v$is the gravitational acceleration.

Equation (1.10) follows the definitions established in Eqs. (1.8) and (1.9), where E_v represents the total energy of the vapor, and q_v^{int} denotes the heat transferred across the interface due to the temperature difference, excluding the latent heat of phase change.

By solving the aforementioned conservation equations, this study will analyze the vapor flow field and heat transfer mechanisms driven by phase change inside the heat pipe.

3.1.5 Heat Pipes verification

In the 2016 book *Heat Pipe Design and Technology* [2], authored by Bahman Zohuri, Chapter 2 provides comprehensive derivations and discussions regarding the various boundary conditions and limiting factors that must be considered when simulating heat pipes. For heat pipes operating under realistic conditions, capillary pressure (Δp_c) is the most common limiting factor. Its formula is expressed as follows:

$$\Delta p_c = 2\frac{\sigma}{r_c} \quad \text{Eq.(1.11)}$$

where σ is the surface tension, and r_c is the effective capillary radius of the porous media region. This pressure is equivalent to the sum of the pressure required to drive the vapor, the pressure drop caused by gravity, and the pressure required to drive the liquid through the porous media region, which is expressed as the following equation:

$$\Delta p_c = \Delta p_v + \Delta p_g + \Delta p_l \quad \text{Eq.(1.12)}$$

According to Darcy's law, we can obtain:

$$\Delta p_l = \frac{\mu_l L_{eff}}{\kappa A_w} \dot{V} \quad \text{Eq.(1.13)}$$

where μ_l is the kinematic viscosity of the liquid, L_{eff} is the effective length of the heat pipe,

κ is the permeability of the porous media region, A_w is the cross-sectional area of the porous media region, and $\dot{V}$ is the volumetric flow rate, given by the following equation:

$$\dot{V} = \frac{\dot{Q}}{\Delta H_{vap}\rho} \quad \text{Eq(1.14)}$$

where $\dot{Q}$ is the heat transfer rate, ρ is the liquid density, and ΔH_{vap} is the latent heat of vaporization. By substituting Eqs. (1.11)-(1.13) into Eq. (1.14), and given that the influence of factors other than the liquid phase can be neglected in most applications, we finally obtain:

$$\dot{Q} = 2\frac{\kappa A_w \Delta H_{vap}\rho\sigma}{\mu_1 L_{eff} r_c} \quad \text{Eq(1.15)}$$

From Eq. (3.15), the maximum heat load capacity of the heat pipe can be calculated. This equation will be utilized for accuracy verification between the simulation model of this study and the practical operation of the heat pipe. Furthermore, the referenced book also delineates the equivalent thermal resistance network diagram for the thermal conduction approximation of a heat pipe, as illustrated in Fig. 3. This diagram will serve as the baseline for calculating the effective thermal conductivity of the heat pipe in this research.

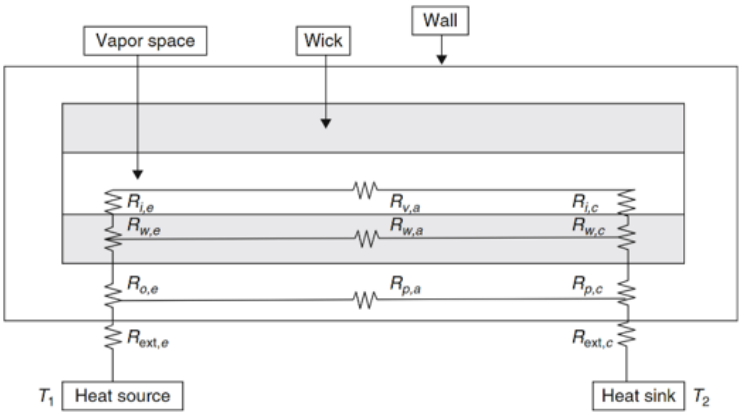


Fig.3: Thermal Resistance Model of the Heat Pipe

3.2 Practical Model:

3.2.1 Cooling fluid decision

Single-phase immersion cooling fluids can be broadly classified by their chemical composition into fluorochemical fluids, synthetic oils, mineral oils, and silicone oils. Due to significant differences in the physical properties of these various coolants, their heat dissipation performance under natural convection and low flow rate conditions varies accordingly. Compared to forced convection systems, natural convection and low-flow cooling rely more heavily on the inherent thermophysical properties of the coolant. Therefore, the selection of the coolant has a crucial impact on the overall heat dissipation effectiveness.

The coolant utilized in this study is the EC-110 Dielectric Coolant, a single-phase immersion fluid developed by Engineered Fluids, USA. Based on the FOM1 (Figure of Merit 1) formula in Eq. (2.1) from the literature, which takes into account the coolant's density,

thermal expansion coefficient, specific heat, thermal conductivity, and kinematic viscosity, the calculated FOM1 value of EC-110 is approximately 51. This indicates that it is a suitable fluid for natural convection. Furthermore, its Global Warming Potential (GWP) is 0, which complies with environmental regulations.

In addition, EC-110 not only possesses extremely high dielectric strength to prevent electronic components from short-circuiting, but it also exhibits chemical stability and excellent material compatibility with heat pipes, heating plates, and other electronic components. This prevents material dissolution or degradation caused by long-term immersion. Consequently, EC-110 is highly ideal as the immersion coolant for this research. Table 1 presents the parameter specifications of EC-110.

$$FOM1 = k(\frac{\beta C_P \rho^2}{\mu k})^{0.2813} \quad \text{Eq(2.1)}$$

k	Thermal conductivity (W/m·K)
β	Thermal expansion coefficient (1/°C)
C_p	Specific heat (kJ/kg·°K)
ρ	Density (kg/m3)
μ	Dynamic viscosity (mPa·s)

Dielectric Strength	>60 kV
Density	820 (kg/m^3)
Kinematic Viscosity	4.44 (cSt)
Thermal Conductivity	0.1363 (W/m·K)
Thermal Expansion Coefficient	0.00067 (1/°K)
Specific Heat	2.2877 (kJ/kg·°K)
GWP	0

Table1. EC-110 properties

3.2.2 Selection of Finned Heat Pipes

This study utilizes commercially available cylindrical heat pipes as the primary heat transfer components. Each heat pipe has a diameter of 8 mm and a length of 200 mm. Based on product specifications, a single heat pipe possesses a maximum heat transfer capacity of approximately 80 W. To ensure sufficient heat dissipation, a total of six heat pipes of identical specifications are used, providing a theoretical maximum total heat processing capacity of 480 W when operating simultaneously. Given that the heating power in this study is set to 100 W, the heat pipe module can effectively transfer the accumulated heat from the coolant to the external environment.

In this system, the heat pipe is divided into an evaporation section and a condensation section.

The evaporation section is submerged in the coolant to absorb the accumulated heat, while the condensation section extends above the coolant surface and is integrated with external cooling fins to dissipate heat into the ambient air. When the coolant absorbs heat generated by the heating plate, this heat is transferred to the evaporation section, causing the internal working fluid to vaporize. The vapor then travels to the condensation section, releases its latent heat, condenses, and returns to the evaporation section via the capillary structure, forming a continuous phase-change heat transfer cycle. Through this mechanism, the heat pipe can rapidly transfer heat out of the coolant without consuming additional power.

To enhance the convective heat exchange between the condensation section and the air, the six heat pipes are divided into three groups, with each group consisting of two heat pipes paired with an aluminum heat sink module. A solid aluminum block measuring 32 mm in length, 32 mm in width, and 40 mm in height serves as the thermal conductor between the heat pipes and the fins. Two 8 mm holes are drilled into the corners of each block to accommodate the heat pipes. Thermal paste is applied inside the holes to minimize thermal contact resistance and ensure optimal heat conduction. Furthermore, aluminum fins, each measuring 32 mm by 40 mm, are attached to the four sides of the aluminum block to maximize the surface area exposed to the air. Since the heat transfer coefficient of air is significantly lower than that of liquid, relying solely on the smooth surface of the heat pipe would result in an insufficient heat dissipation area, causing the temperature of the condensation section to rise and severely impairing the condensation and return flow of the internal working fluid. The addition of fins effectively expands the heat dissipation area, reduces the thermal resistance between the condensation section and the air, and ensures a smooth release of heat into the environment.

3.2.3Internal Agitation Fan

A Delta AFB0412HA fan is installed at the bottom of the coolant tank to serve as an internal agitation fan. With a remarkably low power consumption of only 0.48 W, its primary purpose is not to provide strong forced convection, but rather to induce slight fluid agitation. This ensures that the heat dissipated by the heating plate is evenly distributed throughout the coolant, thereby preventing localized overheating around the heat source. The fan is controlled by an ESP32 microcontroller, utilizing Arduino programming to adjust the output power. The control mechanism relies on Pulse Width Modulation (PWM) to alter the fan's duty cycle. When the PWM duty cycle is high, the average output power increases, resulting in stronger coolant agitation. Conversely, a lower PWM duty cycle yields a lower average output power and correspondingly weaker fluid agitation.

3.2.4 External Forced Convection Fan

An external forced convection fan operating at a fixed power of 3 W is installed outside the condensation section of the heat pipes. Its main function is to enhance the heat exchange capability between the condensation section and the ambient air, ensuring that the heat extracted from the coolant is effectively expelled. If the condensation section fails to dissipate heat adequately, thermal energy will accumulate at the top of the heat pipes, preventing the working fluid from condensing and thereby degrading the circulation efficiency and overall cooling performance. When the fan is activated, the airflow passes over the condensation section and the aluminum fins, carrying away the dissipated heat. Compared to natural convection, forced convection provides a significantly higher convective heat transfer coefficient, which effectively lowers the surface temperature of the condenser and the fins, maintaining a sufficient temperature gradient between the evaporation and condensation sections. This gradient is crucial for driving the continuous evaporation, condensation, and return cycle of the internal working fluid.

Furthermore, the external fan operates at a constant power without any speed adjustment to deliberately eliminate an experimental variable. This design allows the study to focus exclusively on the effects of the heat pipes' presence and the internal agitation fan's power output. By keeping the external cooling conditions consistent across all test scenarios, the risk of external fan speed variations interfering with the experimental data is successfully avoided.

3.2.5 Heating Plate

Due to budget constraints preventing the purchase of an actual CPU as the heat source, this study utilizes a PTC (Positive Temperature Coefficient) heating plate capable of maintaining a stable 100 W output. The heating plate has dimensions of 40 mm by 40 mm, which yields a calculated heat flux of 6.25 W/cm^2. This specific heat flux is sufficient to accurately simulate the thermal load emitted by a modern CPU combined with an Integrated Heat Spreader (IHS).

3.2.6 Data Measurement System

The data measurement system is designed to monitor the surface temperature of the heating plate in real-time and upload the collected data to a cloud platform for subsequent thermal analysis and steady-state determination. The system primarily comprises a thermistor, an ESP32 development board, a Wi-Fi module, and the ThingSpeak cloud platform. A thermistor acts as the primary temperature measurement component and is securely affixed directly to the surface of the heating plate. To minimize thermal contact resistance and guarantee

measurement accuracy, the thermistor is tightly bonded to the surface to eliminate any insulating air gaps. As the plate heats up, the electrical resistance of the thermistor changes in response to the surface temperature variations.

The thermistor is integrated into a voltage divider circuit connected to an analog input pin on the ESP32. The ESP32 reads the resulting voltage signal and utilizes its built-in Analog-to-Digital Converter (ADC) to translate the analog voltage into a digital signal. The software algorithm then converts this voltage value into a corresponding temperature reading based on the thermistor's inherent resistance-temperature characteristics, thereby capturing the surface temperature variations over time. To facilitate long-term data logging and post-processing, the ESP32 leverages its integrated Wi-Fi capabilities to transmit temperature data to the ThingSpeak cloud platform at 5-second intervals. This allows users to remotely monitor real-time temperature fluctuations via a web interface and export the raw data to plot temperature-time curves, which are essential for identifying heating trends and verifying when the system has reached a steady state.

In addition to cloud-based data logging, the system incorporates a WS2812B LED strip for immediate temperature visualization. Controlled by the ESP32, the LED strip dynamically changes color based on the real-time surface temperature, displaying blue for low temperatures, transitioning through green and yellow during the heating phase, and turning red when reaching high temperatures. This visual feedback mechanism allows experimenters to quickly ascertain the system's thermal status without needing to constantly monitor a computer screen or web dashboard.

4. Experiment Setup:

4.1 Practical Model Setup

The experimental setup uses an acrylic container measuring 200 mm × 200 mm × 280 mm as the cooling tank, filled with cooling fluid to a height of 140 mm. A heating plate is placed at the bottom of the tank to simulate the CPU power, while heat pipes serve as the heat transfer components to rapidly extract heat from the inside of the tank to the outside. For the heat exchange mechanism, an internal circulation fan promotes fluid circulation to ensure a uniform coolant temperature. Externally, a forced convection fan is installed at the condenser end of the heat pipes to bring away the heat and dissipate it into the environment.

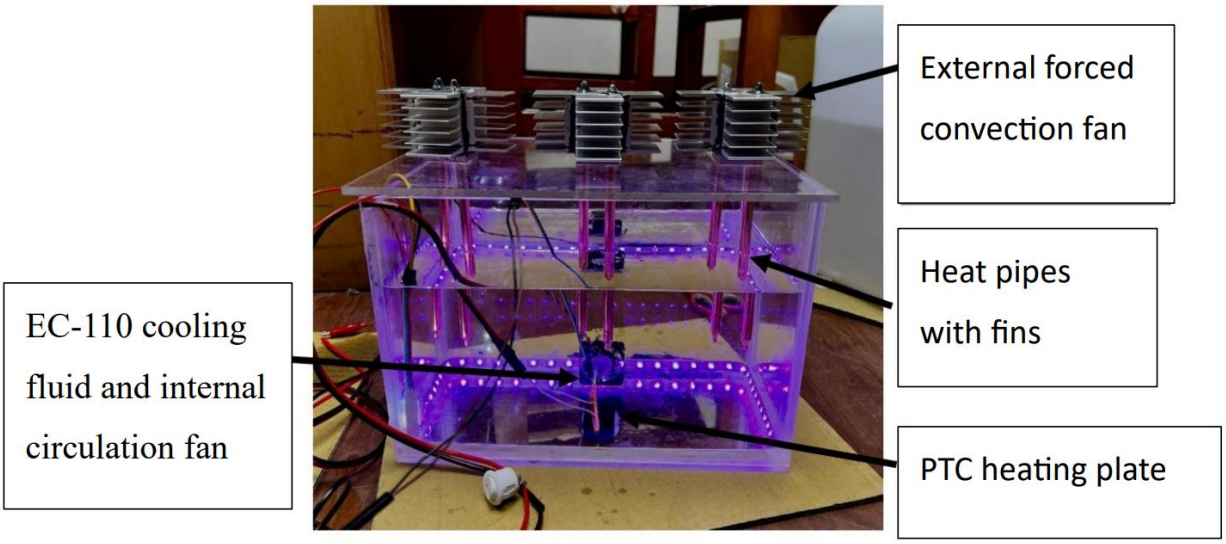


Fig.1 Practical Model Architecture and Structure

4.2 COMSOL Simulation

This research project will utilize COMSOL 6.3 for simulation verification. Model 1 references the thermal conduction approximation described in [6], combined with the equivalent thermal resistance network from [2], to calculate the effective thermal conductivity of the heat pipe for the behavioral simulation of the entire system. To capture more precise heat pipe behavior, Model 2 performs a two-dimensional scale simulation by referencing the LCVF boundary conditions from [6], integrating the capillary limit factors from [2], and utilizing the reference model provided on the COMSOL official website [8]. Within this model, the boundary conditions assume that the vapor inside the vapor chamber is a laminar, compressible ideal gas. Unlike the original LCVF model, the liquid water within the porous media region utilizes the default IAPWS thermodynamic formulation provided by W. Wagner et al. in 1997, which is built into COMSOL 6.3, to accurately represent the temperature-dependent thermophysical properties of water. Furthermore, the Brinkman equations are employed to govern the fluid flow at the interface between the porous media and the vapor chamber flow field, as expressed below:

$$\mathrm{Br} = \frac{\mu u^2}{k(T_w - T_0)} \tag{1.16}$$

Where μ is the dynamic viscosity, u is the flow velocity, k is the thermal conductivity, T_w is the wall temperature, and T_0 represents the fluid temperature. In contrast to the two-phase flow model described in [7], this model implements interface coupling, where the vapor condensing at the interface is converted into an equivalent mass flux of liquid entering the porous media region, and vice versa. This decoupled approach allows the two physical fields to be solved independently, thereby effectively resolving the severe convergence difficulties commonly associated with standard two-phase flow models. Additionally, this model is

capable of predicting whether excessive liquid return resistance will induce dry-out conditions, a critical phenomenon that the conventional LCVF model fails to capture, thereby providing a more comprehensive and realistic simulation of actual operations. The heat pipe has a total length of 150 mm, an outer radius of 10 mm, a wick thickness of 3 mm, and a wall thickness of 1 mm. The geometric configuration corresponds to Fig. 5; from top to bottom, the structure comprises the condensation zone, fins, adiabatic zone, and heat absorption zone, while from the interior to the exterior, it consists of the vapor chamber, the capillary wick layer, and the heat pipe envelope. The material for both the capillary wick matrix and the heat pipe envelope is designated as copper in this study. The external fins are modeled as aluminum, subjected to external forced convection with ambient air at a temperature of 293.15 K and a cross-flow velocity of 3 m/s.

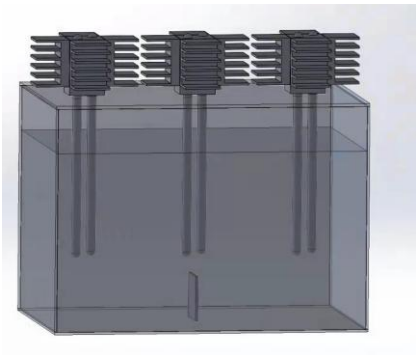


Fig.6： Model 1 Structure

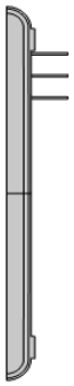


Fig.7： Geometric Structure of the 2D Heat Pipe

The cooling tank designed in this study will adopt the assumptions from [5] to compute the flow field of the coolant after internal fan agitation. These assumptions consider the fluid to be incompressible, possess constant thermophysical properties, operate within a three-dimensional domain, and maintain steady-state flow. The thermophysical properties of the coolant will reference EC-110, developed by Engineered Fluids. This specific coolant boasts a Global Warming Potential (GWP) of zero, which aligns perfectly with this research's vision of mitigating global warming. The computationally efficient RNG k-ε model will be employed as the turbulence model. To compensate for its inherent approximations and to secure a more precise steady-state flow field distribution, the model's convergence tolerance will be strictly set to 10^{-4}. Furthermore, the axial fan model will reference the extreme operating conditions of the Delta-AFB0912VH fan utilized in [4]. While ensuring the product's physical limitations are not exceeded, a prescribed flux will be set for the fan. This setup aims to simulate the thermal dissipation profile of a CPU operating at a power of 100 W and to investigate its maximum surface temperature, ultimately facilitating the evaluation of the system's Power Usage Effectiveness (PUE).

Regarding the geometric modeling and simulation workflow, the design draft of Model 1

will be constructed using the SOLIDWORKS computer-aided design software and subsequently imported into COMSOL Multiphysics 6.3 for advanced multi-physics simulation. Conversely, the design draft of Model 2 will reference the structural configuration in [8], with both the geometric modeling and numerical simulation phases executed entirely within the COMSOL software environment.

5. Methodology:

5.2 Temperature Measure

By using thermistors to monitor the maximum single-point temperature on the chip surface and the 1-minute average coolant temperature in the tank, we evaluate whether the system has reached a steady state. After recording the steady-state coolant temperature, the maximum chip surface temperature, and the fin temperature, we cross-verify these practical model data against the steady-state results calculated by COMSOL under the aforementioned boundary conditions. Finally, we calculate the relative error and the system's PUE.

5.1 Data Measurement and Steady-State Determination

To eliminate instantaneous numerical fluctuations caused by localized fluid convection, this experiment uses a moving-average method as the basis for determining the steady state. The system continuously monitors the maximum single-point temperature on the chip surface T_c and the coolant temperature inside the tank T_w, calculating their moving average rate of temperature change over one minute. The steady-state standard defined in this study is as follows: the system is considered to have reached thermal equilibrium when the variations in both the maximum chip surface temperature and the average coolant temperature over one minute simultaneously satisfy the following condition:

$$\frac{dT}{dt} \leq 0.1(℃/min) \qquad \text{Eq. (3.1)}$$

5.2 Cross-Verification of COMSOL Simulation and Practical Model

The steady-state data obtained from the practical measurements are quantitatively compared with the steady-state simulation results calculated by COMSOL Multiphysics under the same boundary conditions. To evaluate the accuracy of the numerical model, this study employs relative error (ϵ) for analysis. The calculation formula is as follows:

$$\epsilon = \frac{T_{sim} - T_{exp}}{T_{exp}} \times 100\% \qquad \text{Eq. (3.2)}$$

where T_{sim} represents the predicted temperature from the COMSOL simulation, and T_{exp} is the actual thermal equilibrium temperature obtained from practical measurements.

6. Expecting Result

This study conducts both COMSOL simulations and practical prototype experiments. On the experimental side, we obtain real-time temperature rise data by monitoring the thermistors on the heating plate surface. On the simulation side, we ensure solution convergence by configuring fixed-point iterations and pseudo time-stepping. By cross-verifying the measured steady-state temperature against our COMSOL model, we successfully controlled the steady-state error within 5%, satisfying the accuracy requirement of our core validation loop. This result not only confirms that the numerical simulation can accurately predict the fluid and thermal field behaviors of the single-phase immersion cooling system but also establishes a reliable theoretical foundation for subsequent PUE optimization research.

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